

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

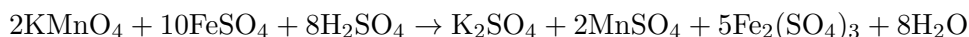
Paper No. CHB-02A: Ancillary Chemistry-I

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) What is glycogen? Where is it stored in the human body?
- (b) What do you mean by expanding octet and incomplete octet? Explain with one example of each.
- (c) In which compounds can the hydrogen atom have the oxidation number -1 ? Explain with two examples.
- (d) Explain the changes in the oxidation numbers of the atoms in the following reaction:



- (e) Differentiate between binding energy and packing fraction.

Question 2.

- (a) What is a spontaneous reaction? Derive the expression for the entropy change (ΔS) of an ideal gas expanding isothermally. [7]
- (b) What is the van 't Hoff reaction isotherm? Derive the Gibbs-Helmholtz equation. [7]

Question 3.

- (a) Differentiate between thin-layer chromatography (TLC) and paper chromatography. [6]
- (b) Describe the analytical applications of the following: (i) Dimethylglyoxime (DMG) and (ii) 8-hydroxyquinoline (oxine). [8]

Question 4.

- (a) What is Fehling's solution? What are the analytical uses of Fehling's solution? [4]
- (b) How would you prepare Benedict's reagent? What happens when Benedict's reagent is treated with glucose in the presence of KSCN? [6]
- (c) Discuss the roles of the following metals in our body: (i) Cobalt and (ii) Zinc. [4]

Question 5.

- (a) Describe the synthesis and uses of the following polymers: (i) Teflon (tetrafluoroethylene) and (ii) Buna-S rubber. [6]
- (b) Describe how iron is extracted from its haematite ore. Give all chemical reactions involved in the blast furnace. ... [6]
- (c) Explain the discrepancy of bond angles in SO_2 and H_2O using VSEPR theory. [2]